

Metagenomics interactive discussion

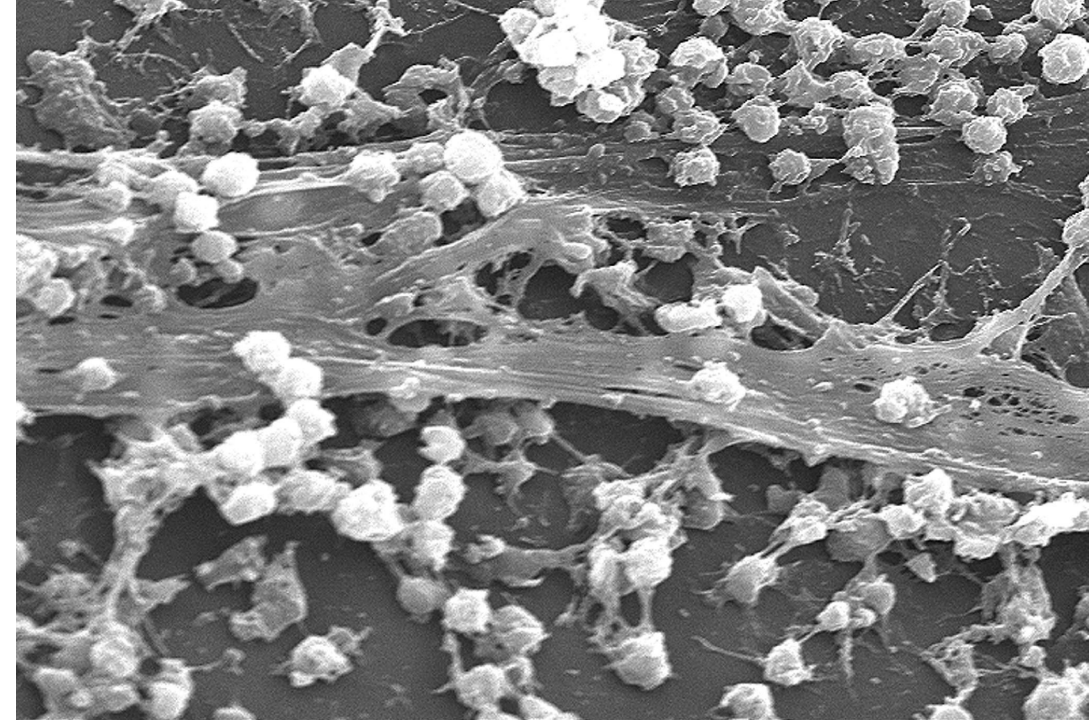
Create a workflow by only using kmer based methods (like mash) to identify a specific strain with high specificity

- Database
- K-mer size
- Extra: distinguish between two very close strains



You have a synthetic community that you use to create a specific biofilm. You want to correlate the abundance composition to the function you measure, but at the same time monitor if your community is not contaminated

- Method
- Reference database
- Potential biases introduced by different methods
- Uncover the potential contaminant



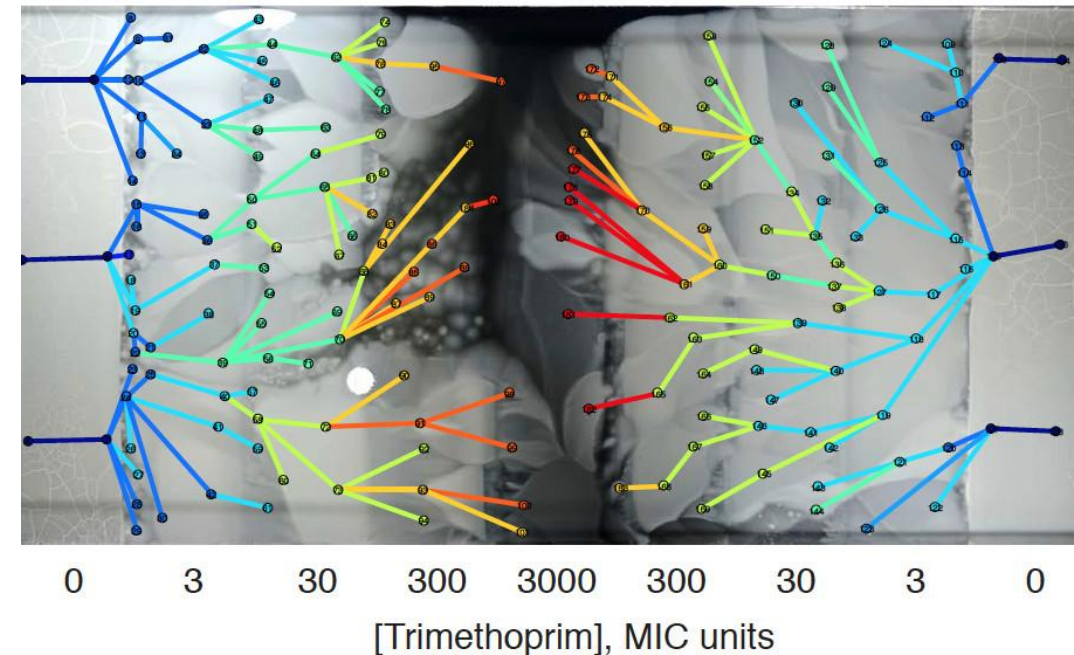
A new study suggests that the presence and specific relative ratio of three different un-culturable bacterial species in the gut show strong correlation with the frequency of migraine. You have a set of 100 unpublished shotgun metagenomics sequencing data from migraine patients, you need to detect these species with high specificity and sensitivity

- Method
- Reference database
- Control experiment
- Intra-species homologies



You are designing a long term (several years) experiment to artificially evolve a bacterial species that only survives in the presence of a well-defined microbial community. You need to monitor the emergence of new gene variants, how do you try to build up the “background” community

- Background community composition
- Detecting the new variants
- Interactions in the community



An industrial wastewater treatment plant relies on anaerobic microbial consortia to digest organic waste into methane biogas. Without warning, Bioreactor 4's methane output drops to zero, volatile fatty acids (VFAs) accumulate, and the reactor crashes. You act as metagenomics consultants equipped with sludge samples from Bioreactor 3 (healthy control) and Bioreactor 4 (failed test).

